

Quick Start Guide: FastAPI Backend

Prerequisites

- Python 3.11+
- All dependencies from `requirements.txt` installed

Installation

```
# Install FastAPI and dependencies  
pip3 install fastapi uvicorn python-multipart
```

Running the Server

Development Mode (with auto-reload)

```
cd /path/to/invoice_app  
uvicorn api_server:app --reload --host 0.0.0.0 --port 8000
```

Production Mode

```
cd /path/to/invoice_app  
uvicorn api_server:app --host 0.0.0.0 --port 8000 --workers 4
```

Background Mode (Linux/Mac)

```
cd /path/to/invoice_app  
nohup uvicorn api_server:app --host 0.0.0.0 --port 8000 > api_server.log 2>&1 &
```

Verify Server is Running

```
curl http://localhost:8000/health
```

Expected response:

```
{
  "status": "healthy",
  "timestamp": "2025-11-24T17:00:00.000000",
  "agents": {
    "tax_agent": "available",
    "bill_agent": "available"
  }
}
```

Access Interactive Documentation

Open your browser and visit:

- **Swagger UI:** <http://localhost:8000/docs>
- **ReDoc:** <http://localhost:8000/redoc>

Test Endpoints

Get Default Bill Invoice

```
curl http://localhost:8000/api/bill/default
```

Process with File Upload

```
curl -X POST http://localhost:8000/api/bill/process \
-F "file=@your_data.xlsx" \
-F "state={\"freight_bill\":{\"runs\":[]}}" \
-F "command=add all details from dataset to invoice"
```

Export to PDF

```
curl -X POST http://localhost:8000/api/bill/export \
-H "Content-Type: application/json" \
-d '{
  "state": {...},
  "format": "pdf"
}' \
--output invoice.pdf
```

Configuration

Change Port

```
uvicorn api_server:app --port 8080
```

Enable CORS for Specific Frontend

Edit `api_server.py`:

```
app.add_middleware(
    CORSMiddleware,
    allow_origins=["http://localhost:3000"], # Your React app URL
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

Troubleshooting

Port Already in Use

```
# Find process using port 8000
lsof -i :8000

# Kill the process
kill -9 <PID>
```

Check Logs

```
# If running in background
tail -f api_server.log

# If running in foreground, logs appear in terminal
```

Module Not Found Errors

Make sure you're in the correct directory:

```
cd /path/to/invoice_app
python3 -c "import invoice_agent_tax; import invoice_agent_bill; print('✓
Modules found')"
```

Next Steps

1. Read the full [API Documentation](#)
2. Test all endpoints using Swagger UI at <http://localhost:8000/docs>
3. Integrate with your React frontend
4. Deploy to production server

Production Deployment

For production, consider using:

- **Gunicorn** with Uvicorn workers
- **Nginx** as reverse proxy
- **Docker** for containerization
- **Environment variables** for configuration

Example production command:

```
gunicorn api_server:app \
  --workers 4 \
  --worker-class uvicorn.workers.UvicornWorker \
  --bind 0.0.0.0:8000 \
  --access-logfile access.log \
  --error-logfile error.log
```